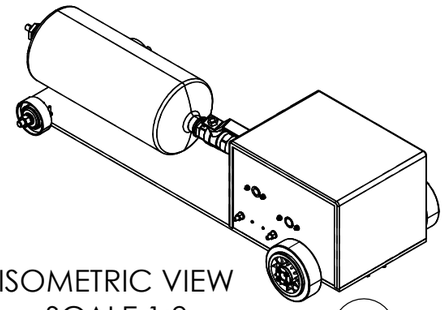
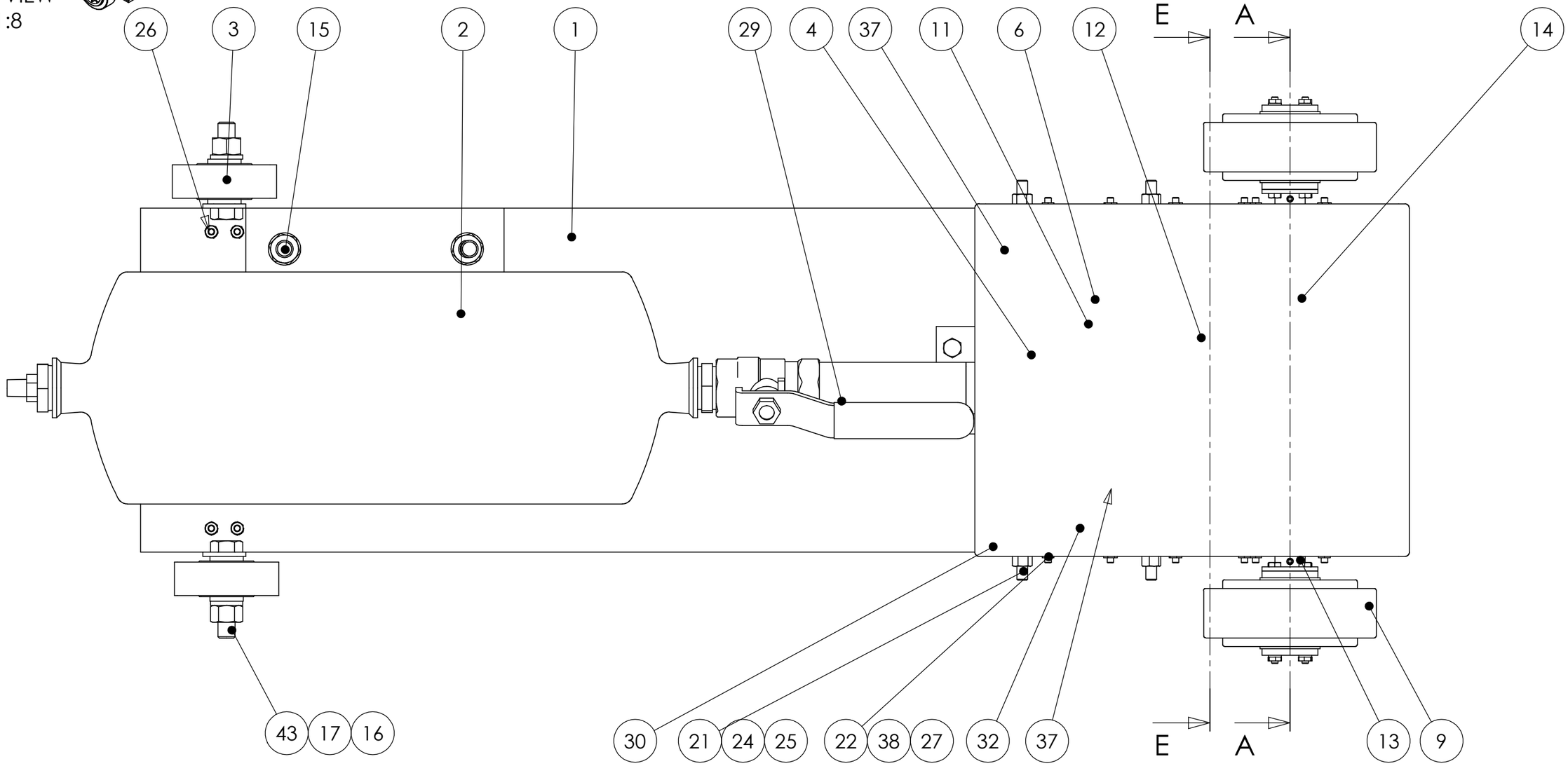


ISOMETRIC VIEW
SCALE 1:8



NOTE: TOP COVER REMOVED
FOR SHOWING TRANSMISSION PARTS CLEARLY

TOLERANCES		THIRD ANGLE PROJECTION
X =± 0.5 X.X =± 0.1 X.XX =± 0.02	ANGULAR ±1° SURFACE FINISH MACHINED FACES Ra 6.3	

	NAME	DATE
DRAWN	YINGFAN GENG	10/25/23
CHECKED	EUGENE LAW	10/25/23
APPROVED	RYAN LEE	10/25/23

MATERIAL: N/A

ALL DIMENSIONS ARE IN MILLIMETRES

DO NOT SCALE DRAWING

A3 SCALE 1:2

TITLE: FULL ASSEMBLY

DWG No. DMF2- 25.001

Imperial College London
Department of Mechanical Engineering

SHEET 1 OF 4 REVISION A

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	DMF-001	BOTTOM PANEL	1
2	DMF-002	AIR VESSEL	1
3	DMF-003	BEARING WHEELS	2
4	DMF-004	TURBINE SUBASSEMBLY	1
5	ISO - Spur gear 2M 36T 20PA 15FW	(RS)SPUR GEAR T36	1
6	ISO - Spur gear 2M 17T 20PA 17FW	(RS)SPUR GEAR T17	1
7	ISO - Spur gear 2M 42T 20PA 15FW	(RS)SPUR GEAR T42	1
8	ISO - Spur gear 2M 17T 20PA 17FW	(RS)SPUR GEAR T17	1
9	DMF-005	BACK WHEELS	2
10	DMF-006	FRONT WHEEL BRACKET	2
11	DMF-007	TURBINE SHAFT	1
12	DMF-008	TRANSMISSION SHAFT	1
13	DMF-009	WHEEL FLANGE	2
14	DMF-010	WHEEL SHAFT	1
15	BS EN 24015 - M8 x 30 x 22-C	HEX BOLT M8 X 30	2
16	BS EN 28738 - 8	PLAIN WASHER M8	4
17	BS EN 24034 - M8 - C	HEX NUT M8	4
18	DMF-011	TOP COVER	1
19	ISO 15 RBB - 1810 - 14,DE,AC,14_68	(RS)DEEP GROOVE BEARING ID 10MM	2
20	ISO 15 RBB - 1812 - 14,DE,AC,14_68	(RS)DEEP GROOVE BEARING ID 12MM	4
21	BS EN 24015 - M5 x 20 x 20-C	HEX BOLT M5 X 20	18
22	BS EN 24015 - M3 x 12 x 12-C	HEX BOLT M3 X 12	12
23	BS EN 24015 - M3 x 45 x 45-C	HEX BOLT M3 X 45	8
24	BS EN 24034 - M5 - C	HEX NUT M5	18
25	BS EN 28738 - 5	PLAIN WASHER M5	18
26	BS EN 24015 - M3 x 20 x 12-C	HEX BOLT M3 X 20	4
27	BS EN 28738 - 3	PLAIN WASHER M3	24
28	BS EN 24035 - M3 - N	HEX NUT M3	4
29	DMF-012	NOZZLE	1
30	DMF-013	SIDE PANEL	2
31	DMF-014	BEARING HOUSING ID12	4
32	DMF-015	BEARING HOUSING ID10	2
33	BS 3673-4 - B019M	(RS)INTERNAL CIRCLIP	2
34	BS 3673-4 - B021M	(RS)INTERNAL CIRCLIP	4
35	BS 3673-4 - S010M	EXTERNAL CIRCLIP	1
36	BS 3673-4 - S012M	EXTERNAL CIRCLIP	2
37	DMF-016	L-BRACKET	2
38	BS EN 24036 - M3 - N	HEX NUT M3	21
39	SPRING PIN ISO 8752 - 3 X 22ST	(RS)SPRING PIN M3 X 22	1
40	SPRING PIN ISO 8752 - 3 X 14ST	(RS)SPRING PIN M3 X 14	2
41	SPRING PIN ISO 8752 - 3 X 26ST	(RS)SPRING PIN M3 X 26	2
42	SPRING PIN ISO 8752 - 3 X 20ST	(RS)SPRING PIN M3 X 20	2
43	BS EN 24015 - M8 x 40 x 40-C	HEX BOLT M8 X 40	2
44	BS EN ISO 8734 - 3 x 8 - A - St	DOWEL PIN M3 X 8	8

TOLERANCES		THIRD ANGLE PROJECTION
X =± 0.5 X.X =± 0.1 X.XX =± 0.02	ANGULAR ±1° SURFACE FINISH MACHINED FACES Ra 6.3	

	NAME	DATE
DRAWN	RYAN LEE	07/11/23
CHECKED	AIDAN LUNG	07/11/23
APPROVED	EUGENE LAW	07/11/23

MATERIAL: N/A

ALL DIMENSIONS ARE IN MILLIMETRES

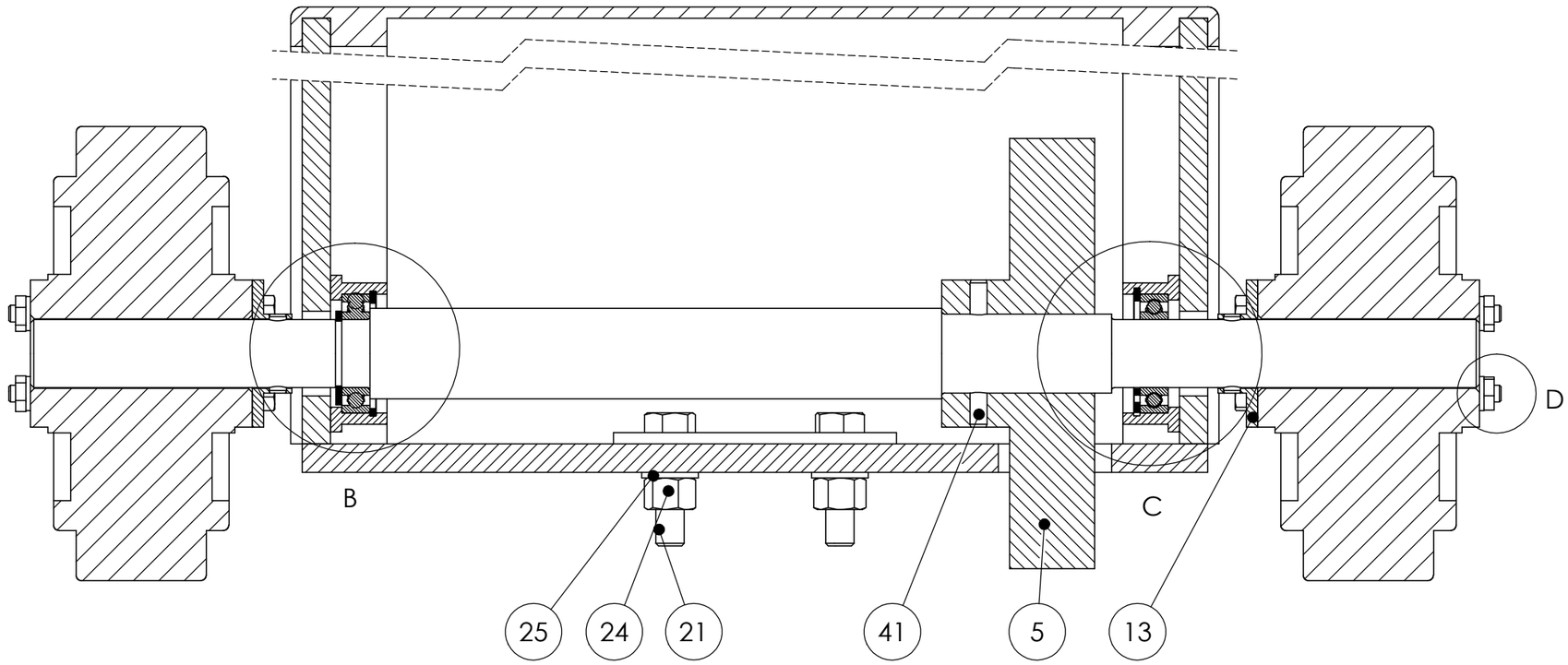
DO NOT SCALE DRAWING

A3 | SCALE 1:2

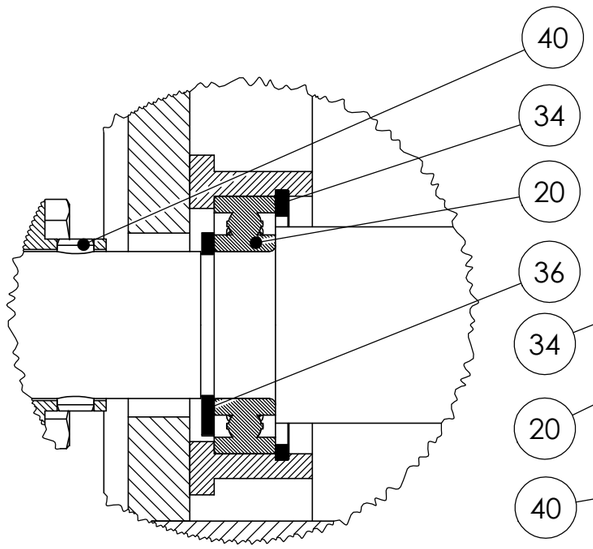
TITLE: BILL OF MATERIALS

DWG No. DMF2-25.001

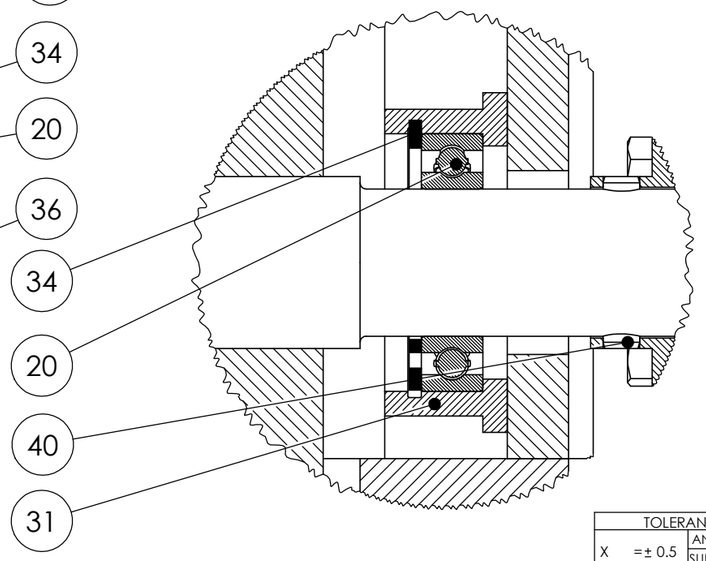
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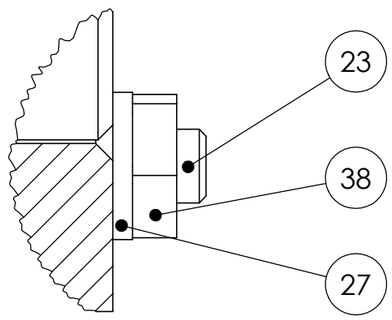
SECTION A-A
SCALE 1 : 1



DETAIL B
SCALE 2 : 1



DETAIL C
SCALE 2 : 1



DETAIL D
SCALE 4 : 1

TOLERANCES		THIRD ANGLE PROJECTION
X =± 0.5 X.X =± 0.1 X.XX =± 0.02	ANGULAR ±1° SURFACE FINISH MACHINED FACES Ra 6.3	

	NAME	DATE
DRAWN	YINGFAN GENG	07/11/23
CHECKED	ERICA GONG	07/11/23
APPROVED	RYAN LEE	07/11/23

MATERIAL: N/A

ALL DIMENSIONS ARE IN MILLIMETRES

DO NOT SCALE DRAWING

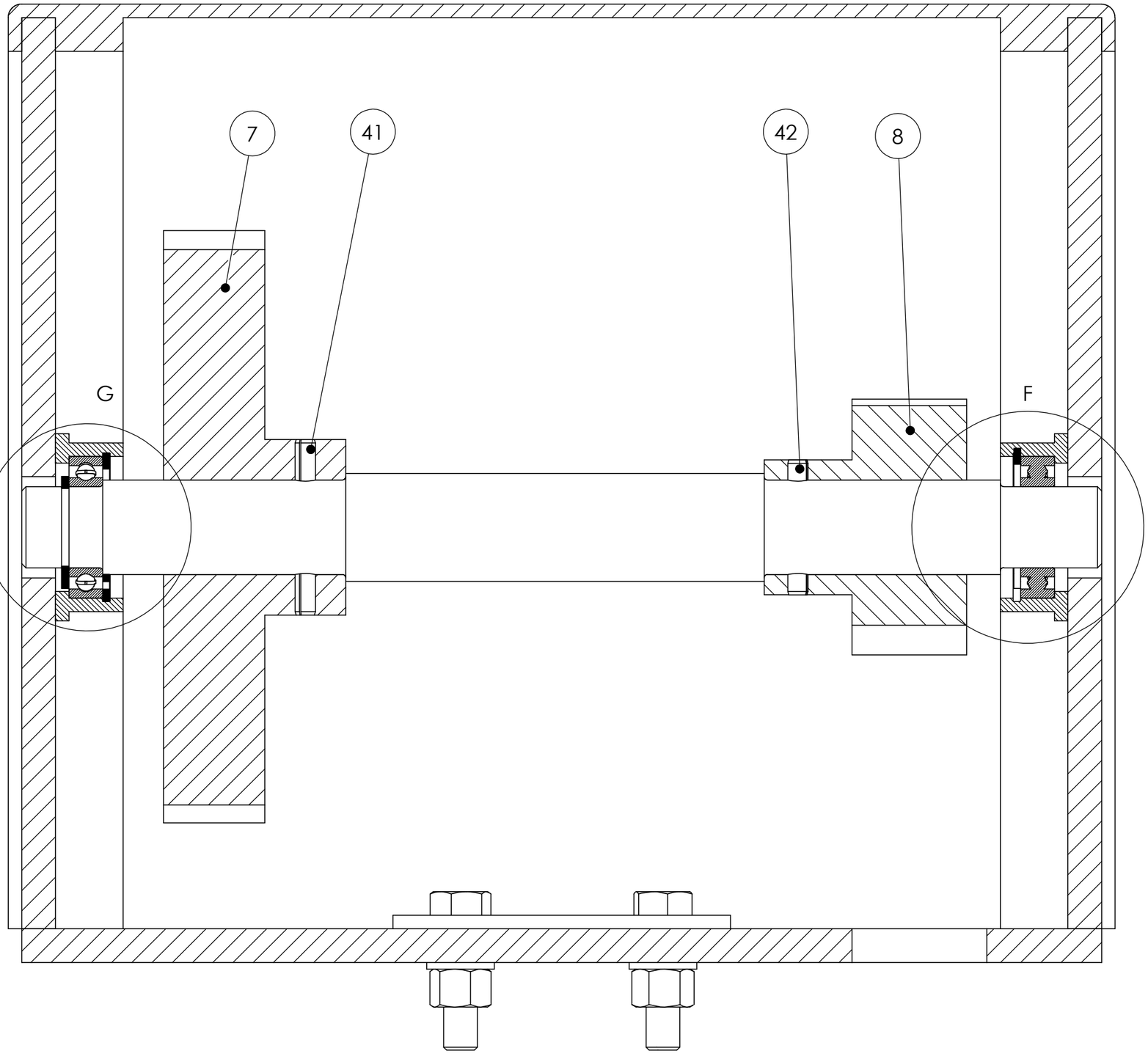
A3 SCALE 1:2

TITLE: FULL ASSEMBLY

DWG No. DMF2-25.001

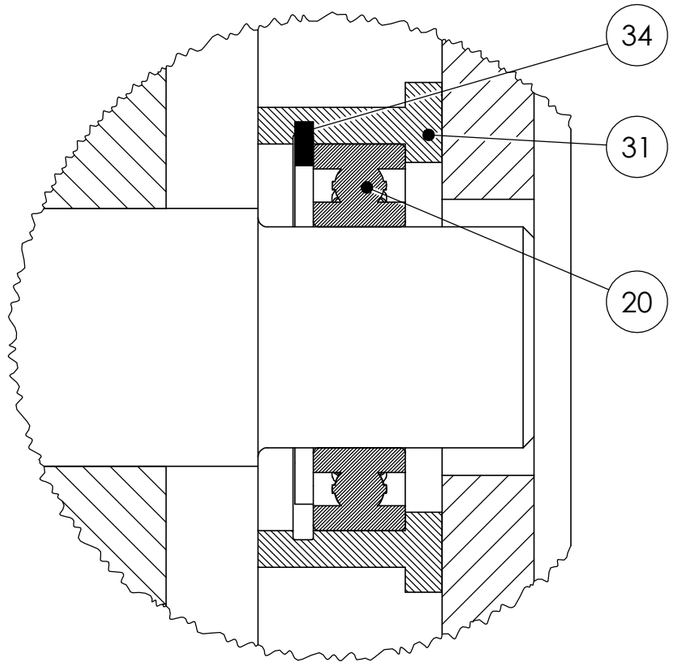
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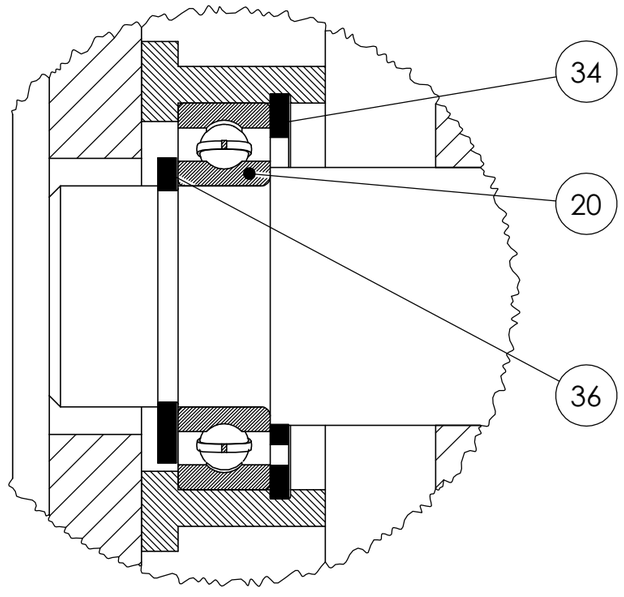


SECTION E-E
SCALE 3 : 2

NOTE: WHEEL AXEL SUBASSEMBLY REMOVED FOR CLEARITY



DETAIL F
SCALE 3 : 1



DETAIL G
SCALE 3 : 1

TOLERANCES		THIRD ANGLE PROJECTION
X =± 0.5 X.X =± 0.1 X.XX =± 0.02	ANGULAR ±1° SURFACE FINISH MACHINED FACES Ra 6.3	

	NAME	DATE
DRAWN	RYAN LEE	07/11/23
CHECKED	AIDAN LUNG	07/11/23
APPROVED	ERICA GONG	07/11/23

MATERIAL: N/A

ALL DIMENSIONS ARE IN MILLIMETRES

DO NOT SCALE DRAWING

A3 SCALE 2:1

TITLE: FULL ASSEMBLY

DWG No. DMF2-25.001

Imperial College London
Department of Mechanical Engineering

SHEET 4 OF 4 REVISION A